TEK5030 Computer Vision
University of Oslo
Project Proposal - Group 5

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We want to use the feed from a stereo camera to do VSLAM in an indoor environment to create a pointcloud. Then we want to make a 3D-model of the environment and determine the gravity vector. When we have a good model of the area, we want to project objects into this area, e.g. a ball that can bounce around using the laws of physics and the estimated geometry of the room to make it look like it bounces as realistic as possible.

Plan:

1. Data Acquisition: Capture stereo video from a RealSense D435 and perform camera calibration.
2. VSLAM Implementation: Use an existing VSLAM framework to map the environment and localize the camera. We will use ORB-SLAM3 because of its stereo support.
3. Gravity Estimation: Determine the "downward" direction without IMU input. We will use a multi-stage approach:
   (a) Initial Estimate: Define the downward vector by holding the camera approximately horizontal at the start of the experiment.
   (b) Plane Fitting: Use RANSAC for plane fitting to identify the floor. This works well for large flat surfaces, but walls and ceilings may be misidentified as the floor. Despite this we think it will be a valuable estimate.
   (c) Line Detection: Use vanishing points from detected lines, assuming most structural lines are either vertical or horizontal, to refine the gravity vector.
   (d) Motion Priors: Assume the primary plane of camera movement is roughly parallel to the horizontal plane.
4. Physics Simulation: Implement 3D object projection and physics (e.g., a bouncing ball or cube) that reacts to the mapped environment.
5. Extension (Optional): If we have time, we will develop a custom VSLAM algorithm to compare against the pre-made solution.